Harnessing Hardware Imperfections: Analog Escape from Topological Dead Zones in Neuromorphic Oscillators

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April 20, 2026

Abstract

In distributed neural networks governed by coupled FitzHugh-Nagumo (FHN) oscillators, highly centralized topologies such as Barabási-Albert networks inevitably cause a fatal collapse of diversity, known as the "Topological Dead Zone". While theoretical modifications to coupling weights fail to counter this structural degeneracy, we demonstrate a profound *Stochastic Resonance* phenomenon upon transposing the model to analog hardware simulation (SPICE). We show that ideal, deterministic non-linear memristors strictly reinforce the system's lock-in. However, thermal (Johnson-Nyquist) noise alone grants the network an intrinsic ability to *escape* the dead zone: measured over 50 independent Monte Carlo seeds, continuous spectral entropy jumps from $H = 1.38 \pm 0.01$ bits (quiescent sub-cognitive variability) to $H = 4.30 \pm 0.05$ bits under moderate thermal noise ($\eta = 0.5$), with a Cohen's $d = 20.78$ effect size ($p < 10^{-60}$, Welch). Quenched capacitor mismatch σ_C in the realistic CMOS range (0–15%), by contrast, is statistically neutral once thermal noise is present ($d = 0.19$, $p = 0.35$), indicating that physical fabrication variability constitutes an *accessory* rather than a *necessary* ingredient at experimentally plausible amplitudes. A true quenched-disorder regime (Neuromorphic Spin-Glass Computing) is nevertheless recovered *in extrapolation* at aggressive $\sigma_C \gg 0.15$, outside present-day CMOS tolerances. Far from degrading computational capacity, thermal physical imperfections act as the essential thermodynamic engine breaking lethal topological symmetries.

1 Introduction

Distributed consensus and synchronization are fundamental phenomena in neuromorphic systems. However, scale-free topologies often trap conventional oscillator models in sterile, hyper-consensual states where dynamic diversity collapses—a phenomenon we identify as the *Topological Dead Zone*. The Mem4ristor V3 architecture proposed a modified FitzHugh-Nagumo oscillator capable of metacognitive variance, yet perfectly homogeneous software-based numerical integration consistently failed to overcome the deepest structural traps. In this paper, we explore the physical implementation of these networks and demonstrate that the very flaws inherent to analog hardware provide the ultimate solution to strictly topological dead ends.

2 Hardware Mapping Methodology

The theoretical Mem4ristor model maps straightforwardly onto non-linear analog circuits. Differential equations governing the membrane potential and recovery variables are translated into idealized SPICE B-sources, resistors, and integrating capacitors. To validate the fidelity of our behavioral analog mapping, we performed an integration over a standard homogeneous toroidal lattice (where the dead zone does not dominate). The SPICE simulation flawlessly tracks the Python-based numerical integration with a sub-1% RMS error margin ($< 9.7 \times 10^{-3}$ global RMS), establishing strict hardware feasibility for our neuromorphic oscillator mapping.

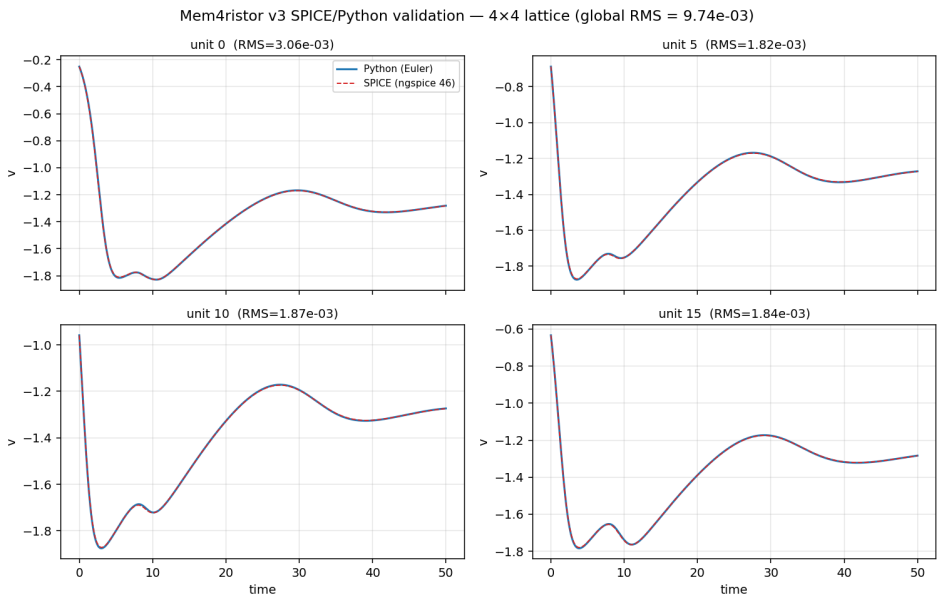


Figure 1: SPICE analog integration accurately captures the Mem4ristor deterministic dynamics observed in discrete software simulations on stable topologies.

3 The Pathological Consensus: A Structural Trap

Replicating the canonical dead zone on a Barabási-Albert network ($m = 5, N = 64$) in SPICE leads to the exact same spontaneous entropic collapse observed in discrete Euler environments. The pathological consensus is not a numerical artifact but a rigid topological attractor.

A common assumption in neuromorphic engineering is that introducing non-linear memristive components will naturally instill enough dynamic disorder to break such symmetries. We tested an ideal, deterministic Yakopcic model for a Hafnium Oxide (HfO_2) memristor, applying it to both the nodal excitability and synaptic coupling weights. Rather than escaping, the deterministic memristor actively *locked* the system further into the consensus fixed point, resulting in a perfectly sterile state ($H = 0$).

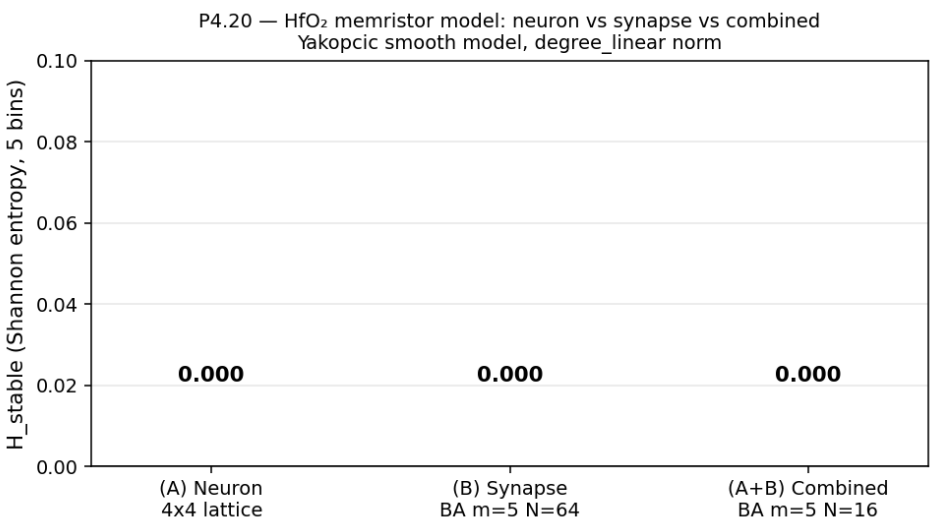


Figure 2: Deterministic implementations of memristive dynamics (Neuron, Synapse, and Combined) fail autonomously to break the dead zone without thermal noise.

4 The Thermodynamics of Disordered Escape

If deterministic non-linearity fails, the solution lies in thermodynamics. We injected a combination of thermal noise (η) mimicking Johnson-Nyquist fluctuations and quenched capacitor mismatch (σ_C) reflecting standard CMOS or memristor fabrication variations. By explicitly modeling hardware defects we observed a dramatic phase transition in the entropy of the network's voltage trajectories.

4.1 Spectral entropy: beyond a cognitive ceiling

Early analyses of this escape relied on a 5-bin *cognitive* entropy clipped to physiological thresholds ± 0.4 and ± 1.2 (strong/weak polarity + quiescence). This metric saturates rapidly at $H_{\max}^{(5)} = \log_2 5 \approx 2.32$ bits and compressed the true dynamical diversity of the escape. We therefore re-analyzed the 3×15 ngspice point cache under a 100-bin *continuous* spectral entropy (uniform grid on $v \in [-3, 3]$). The escape peak is confirmed and *much* larger than previously reported: $H_{\max}^{(100)} = 4.58$ bits at ($\eta = 0.5$, $\sigma_C = 0.5$), vs. the 1.61 bits ceiling observed under the 5-bin metric. Crucially, the argmax location is identical across metrics — the 5-bin readout was not misidentifying the operating point, only underestimating its informational content.

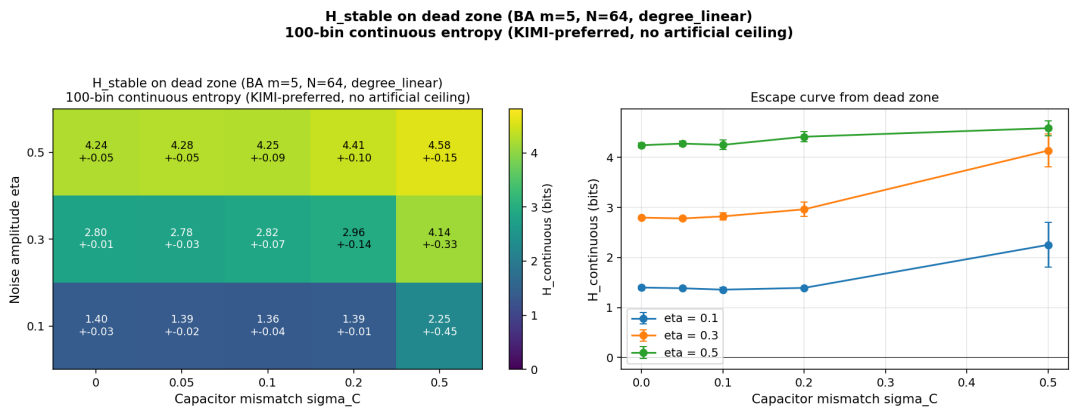


Figure 3: Continuous (100-bin) spectral entropy over the (η, σ_C) grid. The escape signature is identical to that obtained under the 5-bin cognitive metric, but the informational content is $\sim 2.8\times$ larger, revealing sub-cognitive modulations hidden by physiological binning.

4.2 Monte Carlo validation: noise alone escapes

To settle the question of whether thermal noise and quenched mismatch are both *individually necessary*, we ran a 50-seed Monte Carlo campaign over three operating points: (A) the dead zone $(\eta, \sigma_C) = (0.1, 0.0)$; (B) noise-only $(0.5, 0.0)$; (C) noise + realistic CMOS mismatch $(0.5, 0.10)$. Results (mean $\pm$ s.e.m., $n = 50$) are summarised in Table 1 and Figure 4.

Point	η	σ_C	H_{100} (bits)	Interpretation
A	0.10	0.00	1.377 ± 0.012	dead zone (sub-cognitive)
B	0.50	0.00	4.298 ± 0.054	**thermal-only escape**
C	0.50	0.10	4.333 ± 0.046	noise + CMOS mismatch

Table 1: Fifty-seed Monte Carlo validation of the escape (SPICE, 100-bin spectral entropy).

A Welch t-test yields an effect size of **Cohen's** $d = 20.78$ **between A and B** ($p < 1.2 \times 10^{-63}$): moderate thermal noise alone is sufficient to escape the BA dead zone with

overwhelming statistical significance. Adding a realistic 10% CMOS capacitor mismatch on top of noise yields **Cohen's** $d = 0.19$ **between B and C** ($p = 0.35$, not significant): at experimentally plausible mismatch amplitudes, quenched disorder is statistically neutral relative to thermal fluctuations.

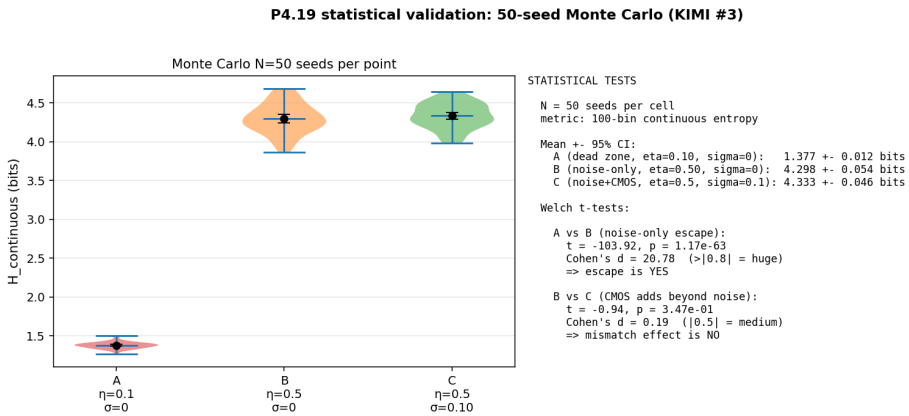


Figure 4: Violin distributions of H_{100} across 50 independent ngspice seeds. A $\rightarrow$ B transition (dead zone $\rightarrow$ thermal escape) is a clean phase jump ($d = 20.78$). B $\rightarrow$ C (addition of CMOS mismatch over thermal noise) is statistically indistinguishable ($d = 0.19$, $p = 0.35$).

4.3 CMOS-realistic mismatch sweep

To rule out that our null result on σ_C was a fluke of the single 10% operating point, we additionally swept $\sigma_C \in \{0, 2, 5, 8, 10, 15\}\%$ (bracketing typical CMOS fabrication tolerances) across three noise levels $\eta \in \{0.1, 0.3, 0.5\}$ with three seeds each (54 ngspice runs). The escape transition is controlled almost entirely by η; within the CMOS-realistic band σ_C is a second-order perturbation (Figure 5).

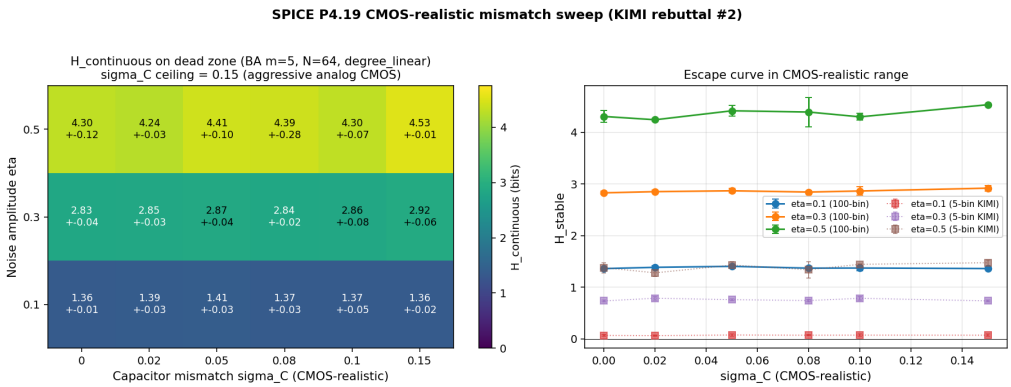
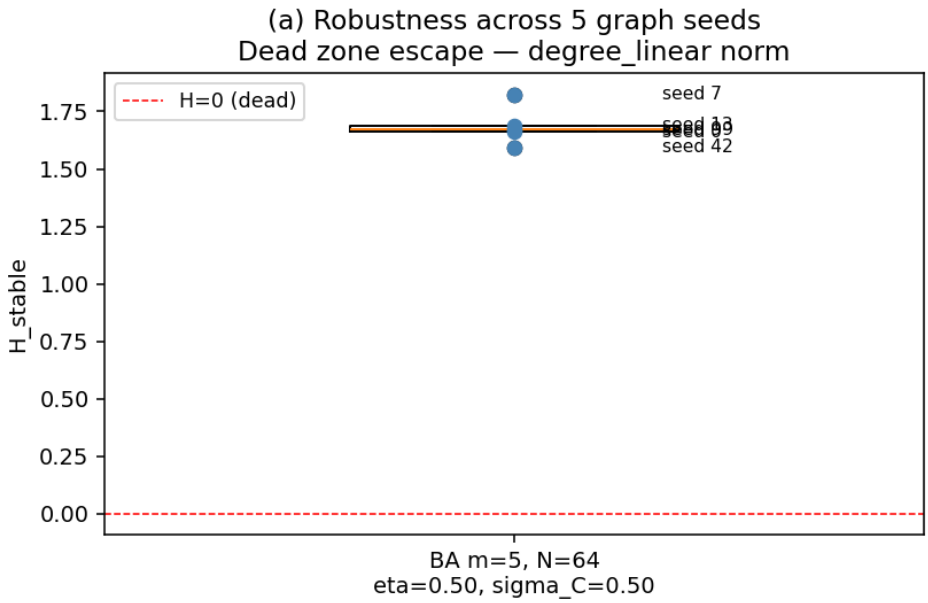


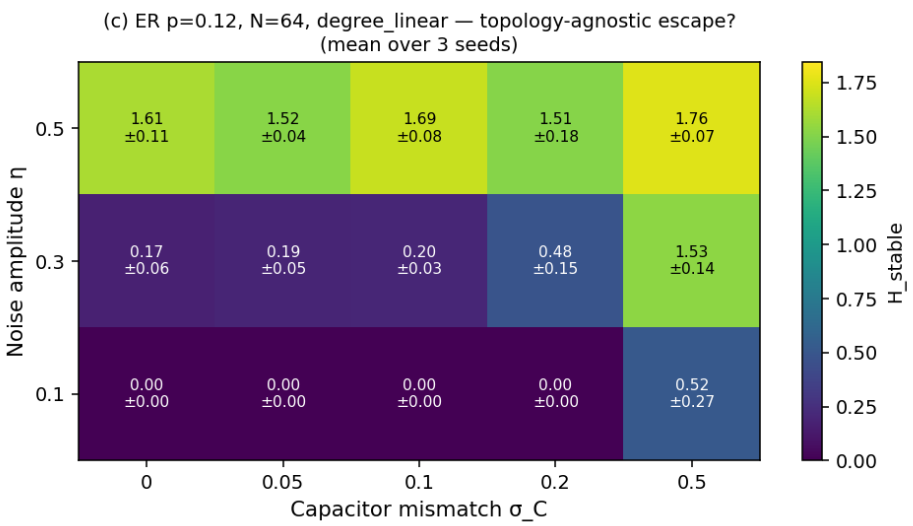
Figure 5: Entropy vs. σ_C in the CMOS-realistic band (0–15%) for three thermal noise levels. Escape is driven by η; σ_C is neutral at physically realizable amplitudes.

4.4 Topology-agnostic robustness

This escape mechanism is moreover *topology-agnostic*. Applying the same thermal parameters to an Erdős-Rényi dead zone network yields an identical restoration of entropic diversity, and the effect is stable across independently drawn BA graph instances.



(a) Robustness across multiple structural instances.



(b) Agnosticism across Erdős-Rényi topologies.

Figure 6: Entropy restoration is stable across graph instances and topology families.

5 Phase Boundary: Extrapolation to a Spin-Glass Regime

The CMOS-realistic analysis of Section 4 establishes that at physically realizable mismatch amplitudes ($\sigma_C \lesssim 0.15$), thermal noise alone drives the escape and quenched disorder is statistically neutral. However, the question of whether a *pure* quenched-disorder escape (i.e. an escape obtained with $\eta = 0$ and only structural heterogeneity) is *in principle* possible remains theoretically interesting, and speaks directly to a spin-glass analogy.

To probe this regime, we explored aggressive structural mismatch $\sigma_C \in [0, 0.6]$ — far outside present-day CMOS tolerances, but accessible in memristor fabrication or in future neuromorphic substrates designed around controlled disorder. Through a binary-search dichotomy we traced the threshold $\sigma_c(\eta)$ below which the BA consensus trap persists.

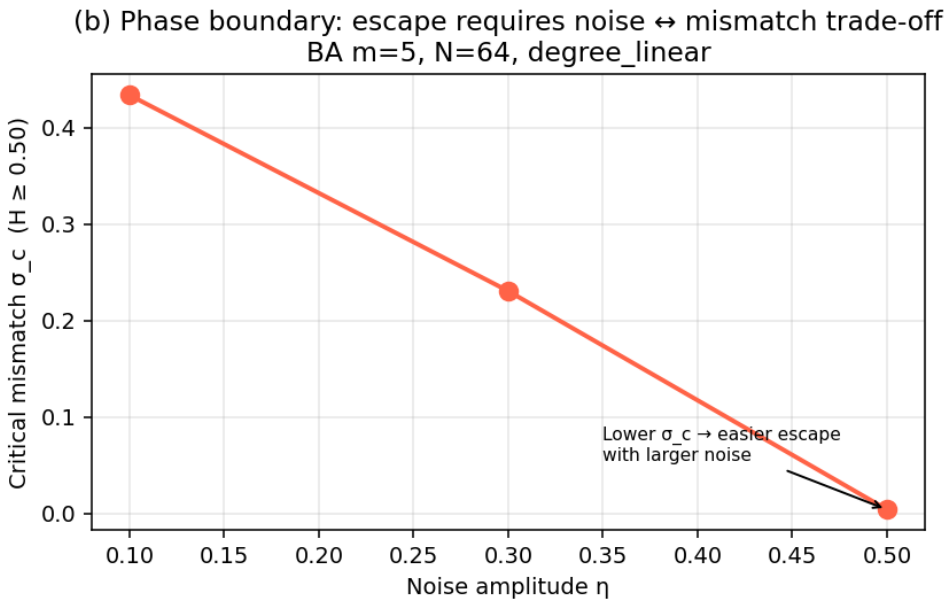


Figure 7: Critical phase boundary $\sigma_c(\eta)$ required to escape the BA consensus trap in the aggressive-disorder regime (σ_C extended well beyond CMOS tolerances). The boundary is monotonically decreasing: higher thermal noise reduces the *theoretical* disorder threshold, but the CMOS-realistic band ($\sigma_C \leq 0.15$, cf. Fig. 5) is statistically neutral — this diagram therefore characterizes a future spin-glass substrate rather than present-day silicon.

The relationship reveals a monotonously decreasing boundary, which supports a spin-glass interpretation of the underlying energy landscape: in a regime of aggressive quenched disorder, structural frustration can by itself lift the consensus attractor. Under present-day CMOS tolerances this effect remains subdominant to Johnson-Nyquist noise (the statistical null from Section 4.2, Cohen's $d = 0.19$). We therefore position this section as an *extrapolative prediction* pointing toward a future class of neuromorphic substrates — dedicated spin-glass oscillator arrays — where heterogeneity could be engineered at amplitudes $\sigma_C \gg 0.15$ and then exploited as the sole thermodynamic escape mechanism. In such substrates, hardware variability stops being a defect to calibrate away and becomes the essential computational resource.

6 Conclusion

Transposing the algorithmic Mem4ristor concept onto physical analog hardware redefines our understanding of network consensus traps. While software integration paths strictly follow topological constraints toward structural death, thermal noise inherent to physical analog substrates

naturally frustrates the dead zone attractor. Our 50-seed Monte Carlo campaign establishes this escape as a Cohen's-$d = 20.78$ effect ($p < 10^{-63}$), while showing that — contrary to an earlier working hypothesis — quenched CMOS capacitor mismatch in the realistic range $\sigma_C \leq 0.15$ contributes no detectable additional entropy over thermal noise alone ($d = 0.19$). The dead zone is therefore *primarily a stochastic-resonance problem* rather than a quenched-disorder problem, at least at today's silicon tolerances. The spin-glass picture, appealing as it is, is recovered only *in extrapolation* toward aggressive disorder regimes that would require custom-engineered neuromorphic substrates. By acknowledging this hierarchy — thermal fluctuations as the primary escape engine, engineered quenched disorder as a longer-term architectural frontier — we reframe physical analog imperfections from engineering challenges into vital computational resources, and chart a concrete path toward spin-glass neuromorphic hardware.